

Using Technology For Community Monitoring In The Indigenous Communities Of Panama

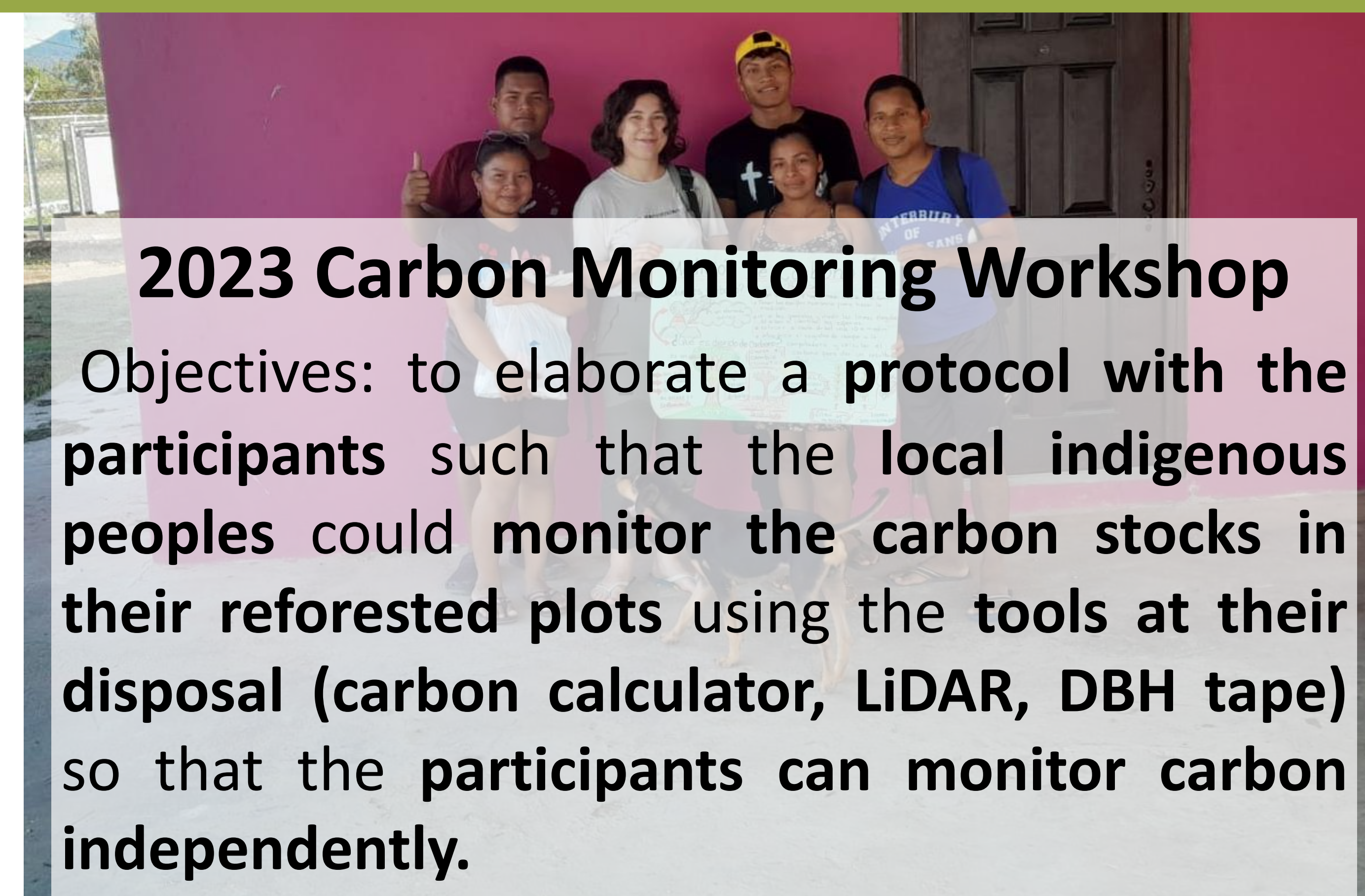
Maximiliane Jousse¹, Madelaine Gauthier¹, Madelaine Chango², Grady Zappone¹, Brais Marchera², Catherine Potvin^{1,2}

1. Department of Biology, McGill University, Montreal, QC, Canada

2. Smithsonian Tropical Research Institute, Panama City, Panama

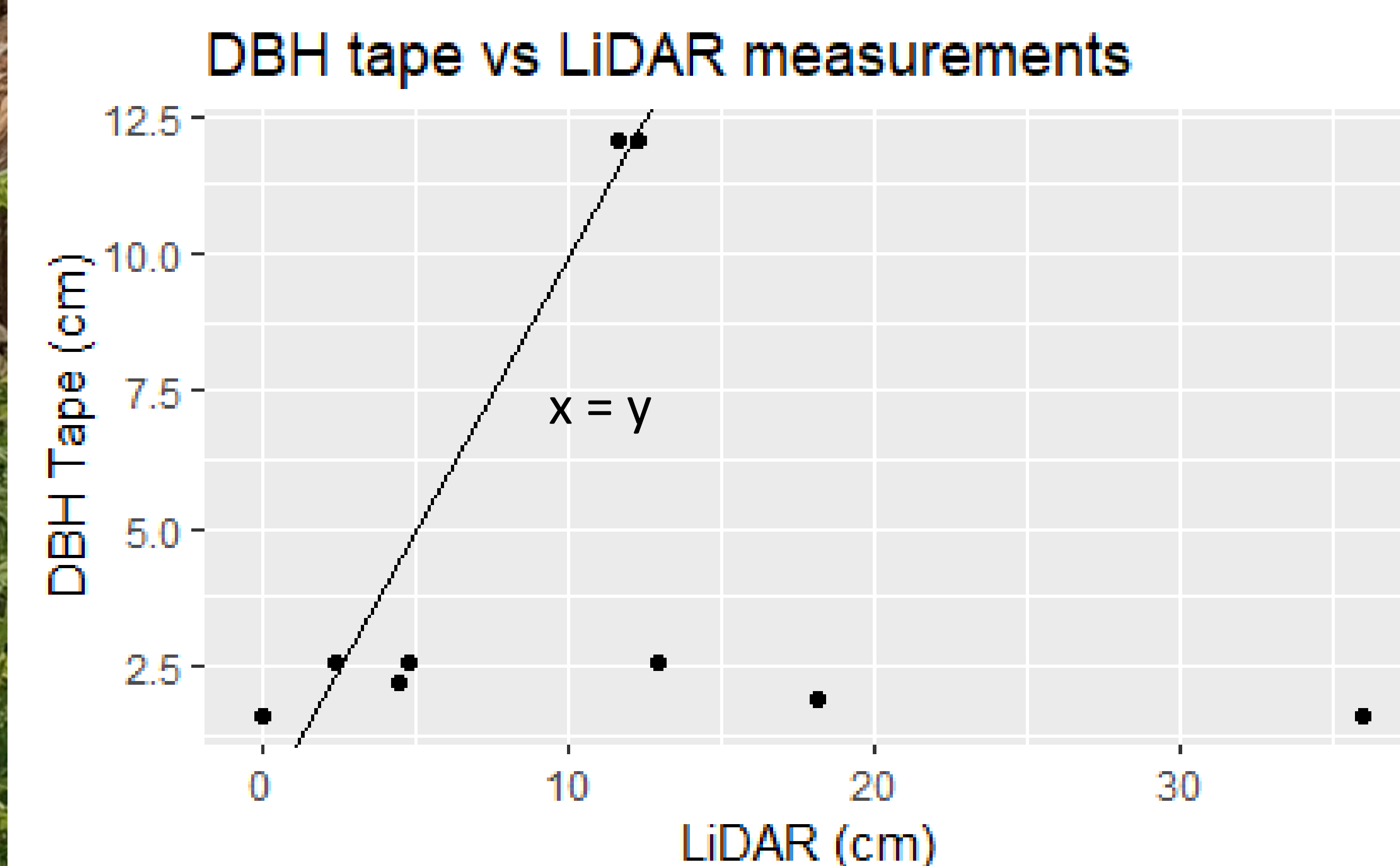
Background

- Tropical forests store about one quarter of all terrestrial carbon and have the potential for re-creating sinks through reforestation.
- The Bayano-McGill Reforestation Project, established in 2020, aims to offset part of the university's carbon emissions through planting and maintaining reforested plots in two Embera communities: Piriati and Ipeti.
- Indigenous communities are stewards, owners or have legally designated rights to about a quarter of the world's global forest estate, more than a third of intact forest landscapes, and almost a fifth of forest carbon stocks.



The Carbon Calculator an excel spreadsheet!

1. Tree height for each tree is estimated from DBH using a regression based on data previously collected in Panama.
2. Above ground biomass is estimated with density (using tree species), DBH and height.
3. Carbon concentration is estimated from above ground biomass.



Results

- Using LiDAR vs DBH tape: DBH tape is 1) more efficient, 2) cheaper, 3) more accurate in tropical forests, and 4) easier to use
- LiDAR garnered interest, and was kept for alternative use such as documenting the reforestation plots
- Teaching technology is necessary for scientific development
- Action engenders action and interest, seen here in reforestation, conservation, and moving towards a green economy.

